



INFORMATICS INSTITUTE OF TECHNOLOGY

Module Code: DOC330

Module Title: Designing Innovative Solutions

Module Leader: Ms. Jananie Mayooreesan

Assessment type: Assignment 1 – Interim Project Report

Group ID: C1

Project Title: DocGPT

Related Industry: Healthcare Industry

Student details:

Student ID	Name
20221804	Muhammed Saad Mazhar
20221803	Dilina Chathuraka Perera
20221802	Rafeek Mohammed Alfar
20221798	Chadithya Dinuwan Udayantha
20222336	Mohamed Zabrin Mohamed Juzail
20222330	Akash Sathiyathan

Abstract

This proposal addresses the significant challenge faced by the healthcare industry in delivering comprehensive and personalized wellness management to individuals. The current systems lack the necessary tools and platforms to empower individuals, provide tailored recommendations, and facilitate effective communication with healthcare providers. To overcome these limitations, this project aims to develop a comprehensive platform that integrates various healthcare solutions, enabling individuals to actively manage their well-being and improve wellness outcomes.

By addressing these key challenges, the proposed solution will empower individuals to take an active role in their well-being, access tailored recommendations, and effectively interact with healthcare providers. This comprehensive platform will bridge the gap between existing healthcare systems and individual needs, leading to enhanced individual empowerment, improved healthcare decision-making, and ultimately, better wellness outcomes.

In conclusion, the project aims to create a personalized wellness platform that empowers individuals, enhances healthcare decision-making, and improves wellness outcomes.

Acknowledgement

We would like to express our gratitude to everyone who has contributed to the successful completion of this report. Firstly, we would like to thank our teachers for their guidance and support throughout the research and writing process. Their constructive feedback and insightful suggestions have been invaluable in shaping this report.

We would also like to extend our appreciation to our seniors who generously shared their time and provided us with the necessary information to conduct this study. Their willingness to participate and their openness in sharing their experiences and insights were instrumental in the development of this report.

Furthermore, we would like to thank our families and friends for their unwavering support, encouragement, and understanding throughout this challenging process.

Finally, we would like to express our appreciation to all the authors of the relevant literature and resources that have contributed to the development of this report. Their knowledge and expertise have been invaluable in guiding my research and writing.

Once again, we express our sincere appreciation to everyone who has contributed to the completion of this report.

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1. Introduction and Description of the Project

1.1. Introduction

The healthcare industry is facing a significant challenge in providing comprehensive and personalized wellness management for individuals. Existing systems often lack the necessary tools and platforms to empower individuals to take an active role in their well-being, access tailored recommendations, and effectively communicate with healthcare providers. This gap in the healthcare landscape hampers the delivery of personalized care and limits the potential for optimal wellness outcomes. To address this challenge, this proposal introduces a project aimed at developing a comprehensive platform that enables individuals to manage their wellness in a personalized manner, access tailored recommendations, and effectively interact with healthcare providers.

1.2. Description

The proposed project seeks to revolutionize the healthcare industry by creating a holistic and integrated healthcare solution. The project will tackle two key aspects that hinder the delivery of personalized care and wellness management. Firstly, it will develop a personalized wellness management platform that considers individual medical history, preferences, and lifestyle choices when providing recommendations and treatment plans. This platform will empower individuals to actively participate in their wellness journey, offering personalized approaches to wellness that cater to their unique needs.

The second aspect of the project involves establishing a robust and secure database management system to integrate electronic health records and other relevant health data. The fragmented nature of health information often poses challenges for healthcare professionals in accessing and utilizing comprehensive patient data for efficient decision-making. By centralizing and securely managing health records, the proposed solution will streamline the retrieval and analysis of patient information, enabling healthcare providers to make informed decisions and deliver personalized care.

2. Problem Specification

2.1. Problems faced by the current system.

Here are some challenges people face in healthcare.

2.1.1. Lack of access to medical care in remote areas

Mapping personal healthcare access and quality worldwide in 2015

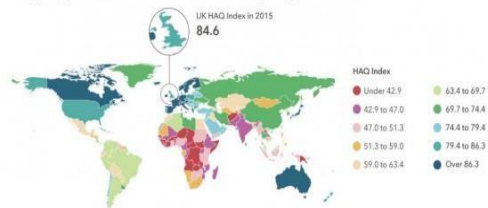


Figure 1 - Lack of access to medical care

The lack of access to medical care in remote areas poses a significant challenge to healthcare delivery. Geographical barriers, limited infrastructure, and a shortage of healthcare professionals make it difficult for individuals residing in remote regions to receive timely and adequate medical attention.

2.1.2. Inefficient communication between healthcare providers.

Inefficient communication between healthcare providers further compounds the problem. Fragmented systems, outdated technologies, and a lack of standardized protocols hinder the seamless exchange of information and coordination of care. This can lead to errors & negatively impacting patient safety and overall healthcare quality.

2.1.3. Inaccurate or incomplete patient data.

The presence of inaccurate or incomplete patient data poses serious risks to effective healthcare. Inaccurate information can lead to misdiagnoses, inappropriate treatments, and compromised patient safety. Furthermore, incomplete patient records limit healthcare providers' ability to make informed decisions, potentially resulting in suboptimal care and compromised outcomes.

2.1.4. High costs of healthcare.

High healthcare costs continue to burden individuals and communities worldwide. The soaring costs of medical procedures, medications, and insurance premiums make healthcare unaffordable for many, leading to delayed or inadequate care.

Percent of adults (age 18 years and older) who reported delaying or going without medical care due to costs, 2021

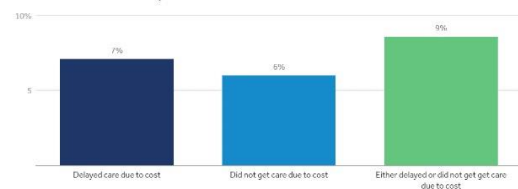


Figure 2 - High cost of living

2.1.5. Absence of a health care chat bot.

The absence of a healthcare chat bot further exacerbates these challenges. A healthcare chat bot could provide immediate access to accurate medical information, assist with symptom triage, and offer preliminary guidance.

2.2. Ideal Solution.

- Create web application that provides healthcare information, remote monitoring capabilities, and appointment scheduling for individuals in remote areas.
- Develop and implement telemedicine platforms that enable remote consultations between patients and healthcare providers.
- Implement comprehensive health record systems that capture and store patient data in a standardized format.
- Develop algorithms and automated processes to validate and ensure the accuracy and completeness of patient data entered into Health Record systems.
- Leverage telehealth technologies and remote monitoring tools to reduce healthcare costs by minimizing the need for physical visits and hospitalizations.
- Develop and deploy an AI-powered chatbot that can provide accurate and reliable medical information, answer common health-related questions, and assist users in identifying appropriate healthcare services.

2.3. Problem Domain

The problem domain would be in the field of healthcare and healthcare technology. Specifically, it encompasses challenges and issues related to medical care access, communication among healthcare providers, patient data management, healthcare costs, and the absence of a medical chatbot.

2.4. Route Causes

1. Lack of access to medical care in remote areas

- Geographical barriers (Remote areas often have limited infrastructure)
- Shortage of healthcare providers (having a shortage of doctors, nurses, and other healthcare professionals)
- Limited resources and funding

2. Inefficient communication between healthcare providers:

- Communication barriers (Healthcare providers may face challenges in terms of time constraints, conflicting schedules, or language barriers)

3. Inaccurate or incomplete patient data

- Lack of standardized data collection
- Fragmented data storage and management (Patient data may be stored in different systems)
- Data entry errors and omissions

4. High costs of healthcare:

- Lack of price transparency
- Expensive medical technologies and treatment

5. Absence of a medical chatbot:

- Lack of technological implementation
- User acceptance and trust (User acceptance and trust in AI-based solutions)

2.5. Stakeholders

- | | |
|--|----------------------------|
| • Patients | • Government |
| • Healthcare Providers (doctors, nurses, specialists, pharmacists) | • Technology Providers |
| • Healthcare Administrators and Managers | • Non-profit organizations |

2.6. Gap Analysis.

	Features	Similar systems (Websites) Locally / Global						Proposed Solution (Idea)
		Sri Lanka Red Cross Society	Lanka Hospitals	Asiri Health	Mount Elizabeth Hospitals	American Hospital Association	Clemenceau Medical Center Dubai	
1	Medical Chat Bot	No	No	No	No	No	No	Innovative
2	Virtual consultations	No	No	No	No	No	No	Innovative
3	Health recommendations	Yes	Yes	Yes	Yes	Yes	Yes	Yes
4	Access to health records	No	Yes	Yes	Yes	No	No	Yes
5	Health education information	Yes	Yes	Yes	Yes	Yes	Yes	Yes
6	Reminders and notification	No	No	No	No	No	No	Innovative
7	Medication management	No	No	No	No	No	No	Innovative
8	Symptom checker	No	No	No	No	No	No	Innovative
9	Wellness program	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10	Language support	No	Yes	Yes	Yes	No	Yes	Yes
11	Emergency assistance	No	No	No	No	No	No	Yes
12	User registration & profiles	No	No	Yes	No	No	Yes	Yes
13	Search Functionality	Yes	Yes	Yes	Yes	Yes	Yes	Yes
14	Find a health care provider	No	No	No	Yes	Yes	Yes	Yes
15	Feedback & rating	No	Yes	Yes	Yes	Yes	Yes	Yes
16	Accessibility feature	Yes	Yes	Yes	Yes	No	Yes	Yes

17	Gamification for health	No	No	No	No	No	No	Innovative
18	Virtual reality therapy	No	No	No	No	No	No	Innovative
19	Social Support Groups	No	No	No	No	No	No	Innovative

Table 1 - Gap Analysis

- **F1 Medical Chat Bot** - AI-powered virtual assistant that provides automated medical information and assistance.
- **F2 Virtual Consultations** - Online video or chat-based consultations with healthcare professionals.
- **F3 Health Recommendations** - Personalized suggestions for improving health and well-being based on user data.
- **F4 Access to Health Records** - Secure online access to personal health records and medical history.
- **F5 Health Education Information** - Comprehensive resources and articles to educate users about various health topics.
- **F6 Reminders and Notifications** - Automated alerts and reminders for medication schedules, appointments, and health milestones.
- **F7 Medication Management** - Tools to track, manage, and receive reminders for medication intake.
- **F8 Symptom Checker** - Interactive tool to assess and provide initial guidance on possible health conditions based on reported symptoms.
- **F9 Wellness Program** - Structured program or activities designed to promote overall well-being and healthy lifestyle habits.
- **F10 Language Support** - Multilingual support to cater to users with diverse language preferences.
- **F11 Emergency Assistance** - Access to emergency services and immediate assistance in critical situations.
- **F12 User Registration & Profiles** - Creation of user accounts and personalized profiles to access specific features and save preferences.

- **F13 Search Functionality** - Ability to search and find specific health information, articles, and resources.
- **F14 Find a Healthcare Provider** - Directory or search functionality to locate nearby healthcare providers based on specialty, location, and user reviews.
- **F15 Feedback & Rating** - User feedback and rating system to provide reviews and ratings for healthcare providers and services.
- **F16 Accessibility Feature** - Inclusion of features to ensure the website is accessible to users with disabilities.
- **F17 Gamification for Health** - Integration of game elements and mechanics to motivate and engage users in adopting healthy behaviors and achieving health-related goals.
- **F18 Virtual Reality Therapy** - Utilization of virtual reality technology to provide immersive and interactive therapeutic experiences for mental health treatment and rehabilitation.
- **F19 Social Support Groups** - Online communities or groups where individuals with similar health concerns or interests can connect, share experiences, and provide emotional support to each other.

3. Solution Outline

3.1. Scope of the Solution

The proposed solution holds a broad scope in addressing various healthcare challenges. By introducing **a website** that incorporates virtual consultations, medical chat bots, access to health records, health education information, reminders, medication management, and more, it aims to revolutionize healthcare delivery. The system caters to remote areas with limited access to medical care, enabling individuals to receive timely and quality healthcare services. It enhances communication and collaboration among healthcare providers, ensuring efficient coordination and improved patient outcomes. Additionally, the system tackles the issues of inaccurate or incomplete patient data, providing a centralized platform for comprehensive and up-to-date health records. With the integration of gamification for health, virtual reality therapy, and social support groups, the solution fosters engagement, mental well-being, and peer support. Overall, the scope of this solution encompasses a holistic approach to healthcare, tackling access, communication, data, and cost challenges while promoting patient empowerment and well-being.

3.2. Aims and Objectives

Aim:

- The aim of the proposed solution is to revolutionize healthcare delivery by addressing key challenges and improving access, communication, data management, and cost-effectiveness.

Objectives:

- Improve Access to Healthcare
- Enhance Communication between Healthcare Providers
- Ensure Accurate and Complete Patient Data
- Reduce Healthcare Costs
- Introduce a Medical Chat Bot

These aims and objectives outline the goals of the proposed solution. By achieving these objectives, the solution aims to transform healthcare delivery and improve overall healthcare outcomes.

4. Key Benefits

These key benefits demonstrate how implementing a system to address healthcare problems can positively impact various stakeholders.

4.1. Patients

- **Enhanced Access** - Patients in remote areas gain access to quality healthcare through this platform, improving their health outcomes and reducing the need for long-distance travel.
- **Timely and Coordinated Care** - Improved communication and data sharing among healthcare providers ensure timely and coordinated care, reducing waiting times.
- **Personalized Care** - Accurate and comprehensive patient data enables healthcare providers to deliver personalized and targeted care, considering patients' medical histories, preferences, and needs.

4.2. Healthcare Providers (Doctors, nurses, specialists, Pharmacists)

- **Efficient Workflow** - Healthcare providers can focus more on patient care and spend less time on paperwork and manual processes.
- **Data-Driven Decision-making** - Access to accurate and comprehensive patient data supports evidence-based decision-making, leading to better diagnoses, treatment planning, and improved patient outcomes.

4.3. Healthcare Administrators and Managers

- **Improved Resource Allocation** - Data analytics and predictive modeling enable efficient allocation of resources, and inventory management thereby reducing costs.
- **Enhanced Operational Efficiency** - Reduced paperwork and improve administrative efficiency.

4.4. Government

- **Cost Containment** - Implementing cost-saving measures and leveraging technology reduces healthcare expenditures.
- **Data for Policy Development** - The system generates valuable data for policymakers.

4.5. Technology Providers

- **Market Opportunities** - The development and implementation of healthcare IT solutions create opportunities for technology providers.

4.6. Non-profit Organizations

- **Enhanced Resource Allocation** - The system's data analytics and reporting features help NPOs make informed decisions about resource allocation.
- **Enhanced Grant and Funding Opportunities** - NPOs that demonstrate effective use of technology solutions and measurable outcomes are more likely to attract grants and funding from foundations.
- **Efficient Volunteer and Resource Management** - The system can streamline volunteer management processes.

5. References

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[Accessed 17 June 2023].

6. Workload Matrix

<i>Tasks</i>	<i>Aaksh Sathiyathan 20222330</i>	<i>Juzail Zabrin 20222336</i>	<i>Chadithya Dinuwan 20221798</i>	<i>Mohammed Alfar 20221802</i>	<i>Dilina Perera 20221803</i>	<i>Saad Mazhar 20221804</i>
Decided a name for the solution which is proposed	✓	✓	✓	✓	✓	✓
Prepared the abstract & acknowledgement		✓			✓	
Prepared the introduction & the description	✓		✓			
Found research articles	✓	✓	✓	✓	✓	✓
Came up with problems faced by people and ideal solutions				✓		✓
Prepared the problem specifications	✓		✓	✓		
Prepared the gap analysis						✓
Wrote the scope of the solution		✓			✓	
Came up with Aims & objectives			✓		✓	
Prepared the key benefits	✓			✓		

Prepared the workload matrix		✓			✓	
Compiled the references in the Harvard referencing format			✓			✓
Checked the grammar of the report	✓					✓
Compilation of the report						✓

Table 2 - Workload Matrix

7. Appendices

1. Artificial intelligence (AI) has found applications in radiology and medical imaging, enhancing their capabilities rather than replacing human involvement. While early expectations of AI as a panacea were unrealistic, the technology has proven valuable in triaging and early detection. AI algorithms have been approved by the U.S. Food and Drug Administration (FDA) for various diseases, with a significant focus on cancer. Doctors at Mass General Brigham utilize around 50 FDA-approved algorithms to aid in patient care, with additional ones undergoing testing. AI enables the swift identification of conditions that require urgent treatment, such as pulmonary embolism, allowing doctors to prioritize patient care effectively. However, it is important to note that AI algorithms do not provide definitive diagnoses but rather indicate the likelihood of specific conditions, guiding physicians' attention. Ultimately, all current levels of AI in healthcare still require human intervention for agreement or disagreement. The integration of AI streamlines healthcare processes and empowers medical professionals to deliver faster and targeted care.

Available at: <https://youtu.be/VePHPymCy2U>

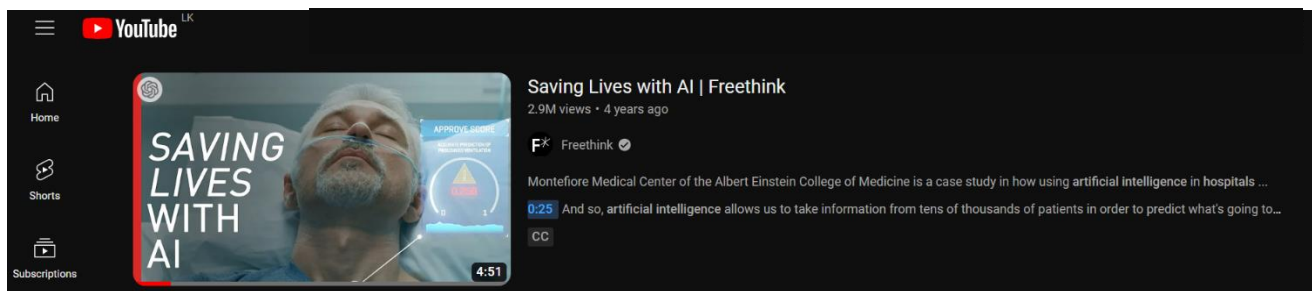


Figure 3 - Saving Lives with AI | Freethink